GCF and LCM

The biggest factor they share. The smallest multiple they share.

Greatest common factor (GCF) of 12 and 18

Factors of 12: 1 2 3 4 6 12

Factors of 18: 1 2 3 6 9 18

1, 2, 3 and 6 are in both lists. The biggest one is 6.

GCF = 6

Least common multiple (LCM) of 4 and 6

Multiples of 4: 4 8 12 16 20 24

Multiples of 6: 6 12 18 24 30 36

Count by each number. 12 is the first one in both lists.

LCM = 12

Use the GCF to pull a sum apart

$$36 + 8 = 4 \times (9 + 2)$$

4 is the GCF of 36 and 8.
36 is 4 × 9 and 8 is 4 × 2.

Your turn!

GCF of 8 and 12 = 4 LCM of 8 and 12 = 24

For teachers

Goal: find the GCF of two numbers up to 100 and the LCM of two numbers up to 12, and use the GCF to pull a sum apart (6.NS.4). Say “List the factors. Which is the biggest one in both?”

GCF and LCM: Practice

Factors for the GCF. Multiples for the LCM.

Part 1 Common factors Ring the factors in both lists. The biggest one is the GCF.

1) Factors of 16: (1) (2) (4) (8) 16
Factors of 24: (1) (2) 3 (4) 6 (8) 12 24
Common factors are ringed. GCF = 8

2) Factors of 15: (1) 3 (5) 15
Factors of 25: (1) (5) 25
Common factors are ringed. GCF = 5

3) Factors of 18: (1) 2 (3) 6 (9) 18
Factors of 27: (1) (3) (9) 27
Common factors are ringed. GCF = 9

4) Factors of 20: (1) (2) 4 (5) (10) 20
Factors of 30: (1) (2) 3 (5) 6 (10) 15 30
Common factors are ringed. GCF = 10

5) Factors of 14: (1) 2 (7) 14
Factors of 21: (1) 3 (7) 21
Common factors are ringed. GCF = 7

6) Factors of 12: (1) (2) 3 (4) 6 12
Factors of 32: (1) (2) (4) 8 16 32
Common factors are ringed. GCF = 4

GCF and LCM: Practice (continued)

Part 2 Find the GCF No help this time.

7) GCF of 24 and 36 = 12

8) GCF of 40 and 56 = 8

9) GCF of 27 and 45 = 9

10) GCF of 32 and 48 = 16

11) GCF of 35 and 63 = 7

12) GCF of 42 and 70 = 14

13) GCF of 36 and 54 = 18

14) GCF of 26 and 39 = 13

15) GCF of 48 and 80 = 16

16) GCF of 21 and 49 = 7

17) GCF of 30 and 75 = 15

18) GCF of 44 and 66 = 22

GCF and LCM: Practice (continued)

Part 2 Find the GCF (continued)

19) **GCF of 18 and 42 = 6**

20) **GCF of 60 and 84 = 12**

21) **GCF of 45 and 72 = 9**

22) **GCF of 28 and 70 = 14**

Part 3 Common multiples Ring the first number in both lists.

23)
Multiples of 3: 3 6 9 **(12)** 15 18 21 24
Multiples of 4: 4 8 **(12)** 16 20 24 28 32
The first one in both lists: LCM = 12

24)
Multiples of 6: 6 12 18 **(24)** 30 36 42 48
Multiples of 8: 8 16 **(24)** 32 40 48 56 64
The first one in both lists: LCM = 24

25)
Multiples of 4: 4 8 12 16 **(20)** 24 28 32
Multiples of 10: 10 **(20)** 30 40 50 60 70 80
The first one in both lists: LCM = 20

26)
Multiples of 5: 5 10 15 20 25 **(30)** 35 40
Multiples of 6: 6 12 18 24 **(30)** 36 42 48
The first one in both lists: LCM = 30

27)
Multiples of 9: 9 18 27 **(36)** 45 54 63 72
Multiples of 12: 12 24 **(36)** 48 60 72 84 96
The first one in both lists: LCM = 36

GCF and LCM: Practice (continued)

Part 3 Common multiples (continued)

28)

Multiples of 6: 6 12 18 24 **(30)** 36 42 48

Multiples of 10: 10 20 **(30)** 40 50 60 70 80

The first one in both lists: LCM = 30

Part 4 Find the LCM No help this time.

29) **LCM of 4 and 6 = 12**

30) **LCM of 3 and 8 = 24**

31) **LCM of 6 and 9 = 18**

32) **LCM of 10 and 12 = 60**

33) **LCM of 5 and 7 = 35**

34) **LCM of 8 and 12 = 24**

35) **LCM of 4 and 9 = 36**

36) **LCM of 6 and 7 = 42**

37) **LCM of 2 and 9 = 18**

38) **LCM of 9 and 10 = 90**

GCF and LCM: Practice (continued)

Part 4 Find the LCM (continued)

39) LCM of 3 and 10 = 30

40) LCM of 8 and 10 = 40

41) LCM of 5 and 12 = 60

42) LCM of 7 and 8 = 56

43) LCM of 11 and 4 = 44

44) LCM of 12 and 9 = 36

Part 5 Pull out the GCF Write the GCF. The numbers in the ( ) have no common factor left.

45) 36 + 8 = 4 × (9 + 2)

46) 45 + 30 = 15 × (3 + 2)

47) 24 + 40 = 8 × (3 + 5)

48) 14 + 35 = 7 × (2 + 5)

49) 54 + 18 = 18 × (3 + 1)

50) 27 + 63 = 9 × (3 + 7)

GCF and LCM: Practice (continued)

Part 5 Pull out the GCF (continued)

51) $20 + 50 = \boxed{10} \times (2 + 5)$

52) $32 + 56 = \boxed{8} \times (4 + 7)$

53) $42 + 12 = \boxed{6} \times (7 + 2)$

54) $64 + 40 = \boxed{8} \times (8 + 5)$

Part 6 More GCF

55) GCF of 18 and 24 = 6

56) GCF of 30 and 42 = 6

57) GCF of 30 and 50 = 10

58) GCF of 33 and 44 = 11

59) GCF of 18 and 48 = 6

60) GCF of 15 and 21 = 3

61) GCF of 24 and 30 = 6

62) GCF of 6 and 8 = 2

GCF and LCM: Practice (continued)

Part 6 More GCF (continued)

63) GCF of 22 and 55 = 11

64) GCF of 22 and 33 = 11

65) GCF of 30 and 35 = 5

66) GCF of 36 and 63 = 9

67) GCF of 28 and 32 = 4

68) GCF of 40 and 50 = 10

69) GCF of 25 and 40 = 5

70) GCF of 4 and 6 = 2

71) GCF of 32 and 40 = 8

72) GCF of 35 and 56 = 7

GCF and LCM: Practice (continued)

Part 7 More LCM

73) LCM of 5 and 11 = 55

74) LCM of 2 and 6 = 6

75) LCM of 7 and 10 = 70

76) LCM of 6 and 12 = 12

77) LCM of 7 and 9 = 63

78) LCM of 3 and 4 = 12

79) LCM of 4 and 10 = 20

80) LCM of 3 and 7 = 21

81) LCM of 6 and 8 = 24

82) LCM of 10 and 11 = 110

83) LCM of 9 and 11 = 99

84) LCM of 6 and 10 = 30

GCF and LCM: Practice (continued)

Part 7 More LCM (continued)

85) LCM of 2 and 5 = 10

86) LCM of 2 and 7 = 14

Part 8 More pulling out the GCF

87) 72 + 63 = 9 × (8 + 7)

88) 12 + 54 = 6 × (2 + 9)

89) 21 + 6 = 3 × (7 + 2)

90) 14 + 2 = 2 × (7 + 1)

91) 36 + 81 = 9 × (4 + 9)

92) 54 + 30 = 6 × (9 + 5)

93) 18 + 16 = 2 × (9 + 8)

94) 56 + 32 = 8 × (7 + 4)

95) 81 + 9 = 9 × (9 + 1)

96) 15 + 27 = 3 × (5 + 9)

GCF and LCM: Practice (continued)

Part 9 Remember? Problems from earlier packets.

97) $2.6 \times 1.5 =$ **3.9**

98) 60% of 45 = **27**

99) $\frac{4}{7} \div \frac{1}{14} =$ **8**

100) 5,688 ÷ 24 = **237**

101) 8 : 5 = 32 : **20**

102) 3.92 ÷ 0.7 = **5.6**

Part 10 Word problems Factors or multiples? Read carefully.

103) Hot dogs come in packs of 8 and buns in packs of 6. What is the fewest of each so none are left?

Number sentence **LCM of 8 and 6 = 24**

Answer **24** of each

Draw or show your work

104) 24 apples and 36 oranges go into the same number of bags, all alike. What is the most bags?

Number sentence **GCF of 24 and 36 = 12**

Answer **12** bags

Draw or show your work

105) One bus comes every 10 minutes, another every 15. Both just came. In how many minutes again?

Number sentence **LCM of 10 and 15 = 30**

Answer **30** minutes

Draw or show your work

GCF and LCM: Practice (continued)

Part 10 Word problems (continued)

106) A 42 cm by 56 cm board is cut into the biggest squares with no waste. How long is a side?

Number sentence **GCF of 42 and 56 = 14**

Answer **14** cm

Draw or show your work

107) Challenge: 18 boys and 27 girls make teams, all alike. What is the most teams? How many girls each?

Number sentence **GCF = 9 teams, 27 ÷ 9 = 3**

Answer **3** girls

Draw or show your work

Check-In: GCF and LCM

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Common factors

1) Factors of 18: (1) (2) (3) (6) 9 18

Factors of 30: (1) (2) (3) 5 (6) 10 15 30

Common factors are ringed. GCF = 6

Part 2 Find the GCF

2) **GCF of 33 and 55 =** 11

3) **GCF of 36 and 60 =** 12

4) **GCF of 28 and 42 =** 14

Part 3 Find the LCM

5) **LCM of 12 and 8 =** 24

6) **LCM of 6 and 15 =** 30

7) **LCM of 9 and 6 =** 18

Part 4 Pull out the GCF

8) **30 + 45 =** 15 **× (2 + 3)**

9) **48 + 18 =** 6 **× (8 + 3)**